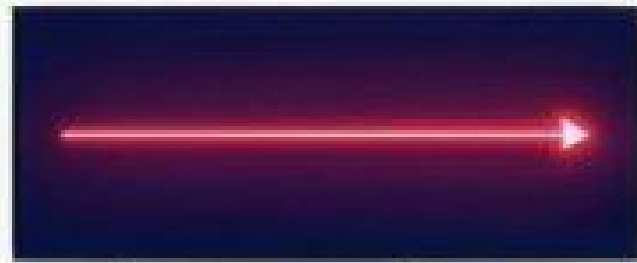
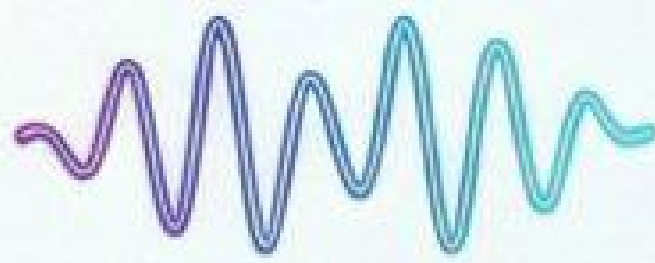


TWO VIEWS OF LIGHT: THE BRANCHES OF OPTICS



RAY (GEOMETRICAL) OPTICS
Treats light as a straight ray.
Explains reflection, refraction.



WAVE OPTICS
Treats light as a wave. Explains
interference, diffraction, polarization.

HISTORICAL THEORIES OF LIGHT



Newton's Corpuscular Theory
Proposed that light is composed of tiny,
perfectly elastic particles called

SUCCESSSES & FAILURES

Success: Reflection, retraction.
Failure: Interference, diffraction,
polarization.



Huygens' Wave Theory
Light is a wave propagating through
an ether medium.

SUCCESSSES & FAILURES

Success: Reflection, refraction,
interference, diffraction, polarization.
Failure: Photoelectric effect, light
speed in denser media.

A Visual Guide to Wave Optics

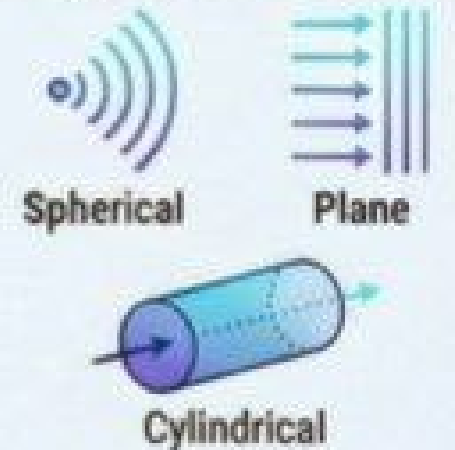
HUYGENS' PRINCIPLE: HOW LIGHT WAVES TRAVEL

What is a Wavefront?
Locus of all points oscillating
in the same phase.

DIRECTION
OF LIGHT

Step 1: Secondary Wavelets

3 Types of Wavefronts



INTERFERENCE: WHEN LIGHT WAVES MEET

Superposition Principle: Resultant displacement is the vector sum of individual displacements.

What is Interference?
Modification in light
energy distribution
when coherent waves
waves superpose.

CONSTRUCTIVE



DESTRUCTIVE

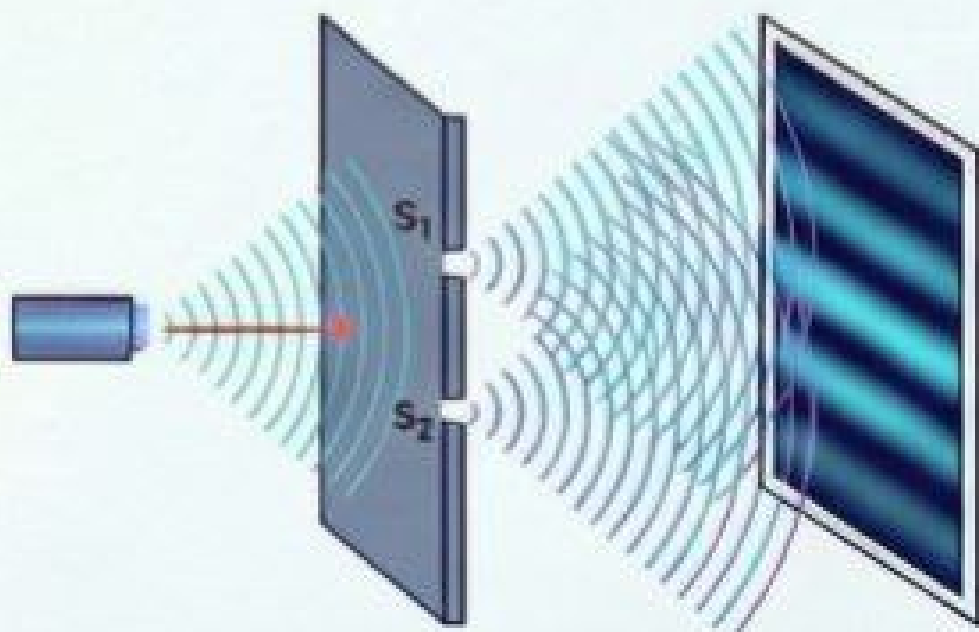


CONDITIONS FOR INTERFERENCE

Type of interference	Path Difference (Δx)	Phase Difference ($\Delta \phi$)
Constructive	$n\lambda$	$2n\pi$
Destructive	$(2n + 1)\lambda/2$	$(2n + 1)\pi$

($n = 0, 1, 2, 3, \dots$)

YOUNG'S DOUBLE SLIT EXPERIMENT (YDSE)



The First Proof of Light's Wave Nature

Alternating Bright and Dark Fringes
Predictable pattern of bright (constructive) and
dark (destructive) bands.



Fringe Width (β)

$$\beta = \frac{\lambda D}{d}$$

λ : wavelength

D : screen-slit distance

d : slit separation

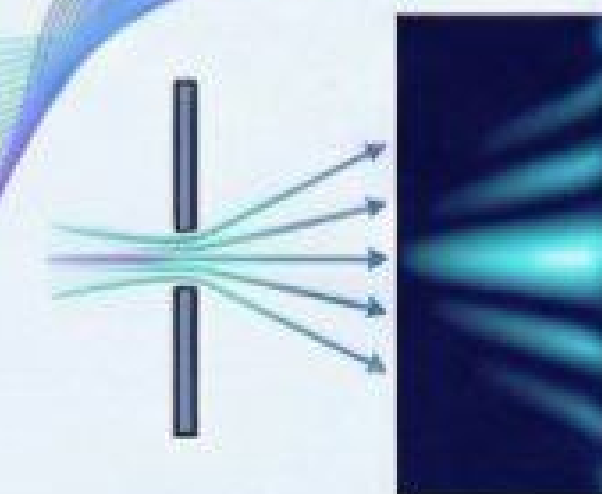
Energy is Redistributed, Not Created

The dark fringes are dark because their light
energy has been redistributed to the bright
fringes, making them extra bright. This adheres
to the law of conservation of energy.

Coherent Sources are Essential

The experiment requires coherent sources,
which emit light of the same frequency with a
constant phase difference. This is achieved by
splitting light from a single source into two.

DIFFRACTION: WHEN LIGHT BENDS



Bending Around Corners:

Light bends around obstacles, entering
the shadow region.

When is Diffraction Noticeable?

Most prominent when obstacle size is
comparable to wavelength (λ).

Diffraction Pattern vs. YDSE Pattern

Unlike uniform YDSE fringes, diffraction
has a very bright, wide central maximum
with much weaker secondary maxima.